

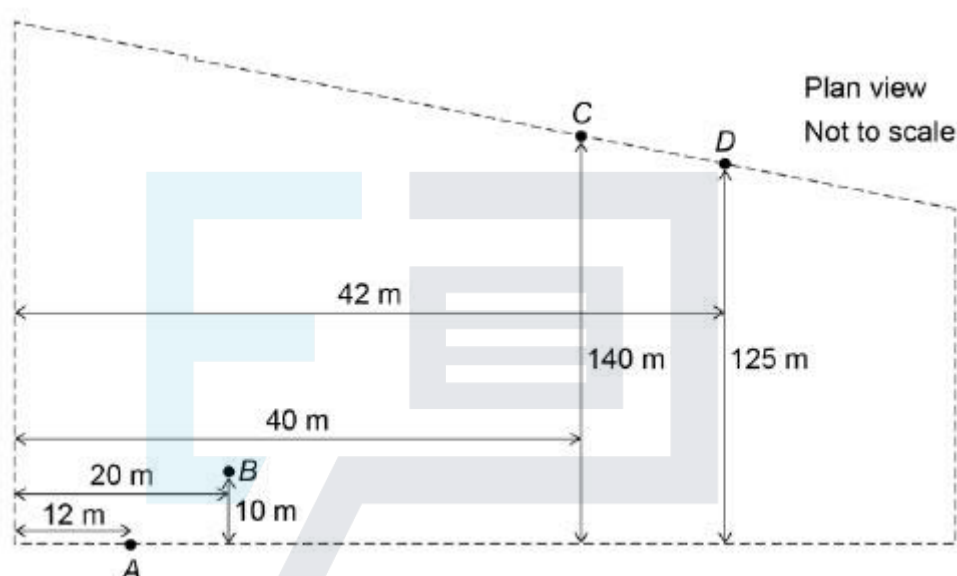
Q1.

A lighting engineer is setting up part of a display inside a large building. The diagram shows a plan view of the area in which he is working.

He has two lights, which project narrow beams of light.

One is set up at a point 3 metres above the point A and the beam from this light hits the wall 23 metres above the point D .

The other is set up 1 metre above the point B and the beam from this light hits the wall 29 metres above the point C .



- (a) By creating a suitable model, show that the beams of light intersect. (6)
- (b) Find the angle between the two beams of light. (3)
- (c) State one way in which the model you created in part (a) could be refined. (1)

(Total 10 marks)

Q2.

- (a) The points A , B and C have coordinates $(3, 1, -6)$, $(5, -2, 0)$ and $(8, -4, -6)$ respectively.

- (i) Show that the vector $\overrightarrow{AC}$ is given by $\overrightarrow{AC} = n \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$, where n is an integer.

(1)

(ii) Show that the acute angle ACB is given by $\cos^{-1}\left(\frac{5\sqrt{2}}{14}\right)$. (4)

(b) Find a vector equation of the line AC . (2)

(c) The point D has coordinates $(6, -1, p)$. It is given that the lines AC and BD intersect.

(i) Find the value of p . (4)

(ii) Show that $ABCD$ is a rhombus, and state the length of each of its sides. (4)
(Total 15 marks)

Q3.

The points A , B and C have coordinates $(3, -2, 4)$, $(1, -5, 6)$ and $(-4, 5, -1)$ respectively.

The line l passes through A and has equation $\mathbf{r} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} 7 \\ -7 \\ 5 \end{bmatrix}$.

(a) Show that the point C lies on the line l . (2)

(b) Find a vector equation of the line that passes through points A and B . (3)

(c) The point D lies on the line through A and B such that the angle CDA is a right angle.

Find the coordinates of D . (5)

(d) The point E lies on the line through A and B such that the area of triangle ACE is three times the area of triangle ACD .

Find the coordinates of the two possible positions of E . (4)
(Total 14 marks)

Q4.

The points A and B have coordinates $(4, -2, 3)$ and $(2, 0, -1)$ respectively.

The line l passes through A and has equation $\mathbf{r} = \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix}$.

- (a) (i) Find the vector $\overrightarrow{AB}$.

(2)

- (ii) Find the acute angle between AB and the line l , giving your answer to the nearest degree.

(4)

- (b) The point C lies on the line l such that the angle ABC is a right angle. Given that $ABCD$ is a rectangle, find the coordinates of the point D .

(6)

(Total 12 marks)

Q5.

The planes Π_1 and Π_2 have equations

$$\mathbf{r} \cdot \begin{bmatrix} 2 \\ 1 \\ 7 \end{bmatrix} = 10 \quad \text{and} \quad \mathbf{r} \cdot \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix} = 7$$

respectively.

- (a) Determine, to the nearest degree, the acute angle between Π_1 and Π_2 .

(4)

- (b) By setting $z = t$, find cartesian equations for the line of intersection of Π_1 and Π_2 in the form

$$\frac{x-a}{l} = \frac{y-b}{m} = z=t$$

(6)

- (c) The line L , with equation $\mathbf{r} = \begin{bmatrix} 20 \\ -1 \\ 7 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 9 \\ 4 \end{bmatrix}$, intersects Π_1 at the point P and Π_2 at the point Q .

Show that $PQ = k\sqrt{2}$, where k is an integer.

(6)

(Total 16 marks)

Q6.

The line l_1 has equation $\mathbf{r} = \begin{bmatrix} 0 \\ -2 \\ q \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$, where q is an integer.

The line l_2 has equation $\mathbf{r} = \begin{bmatrix} 8 \\ 3 \\ 5 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}$.

The lines l_1 and l_2 intersect at the point P .

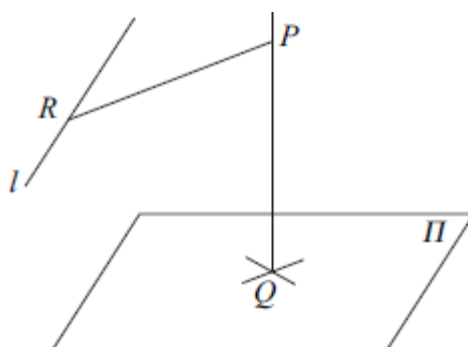
- (a) Show that $q = 4$ and find the coordinates of P . (3)
- (b) Show that l_1 and l_2 are perpendicular. (1)
- (c) The point A lies on the line l_1 where $\lambda = 1$.
- (i) Find AP^2 . (2)
- (ii) The point B lies on the line l_2 so that the right-angled triangle APB is isosceles.
- Find the coordinates of the two possible positions of B . (6)

(Total 12 marks)

Q7.

The point Q has position vector $\mathbf{q} = \begin{bmatrix} 7 \\ 4 \\ 6 \end{bmatrix}$, the plane Π has equation $\mathbf{r} \cdot \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = 36$,
 and the line l has equation $\mathbf{r} = \begin{bmatrix} 20 \\ -8 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} -7 \\ 5 \\ 3 \end{bmatrix}$.

- (a) Show that Q lies in Π . (1)
- (b) Show also that l is parallel to Π . (2)
- (c) The diagram shows the point P , which lies on the normal to Π that passes through Q .
 The point R is the point on l which is closest to P , and $PQ = PR$.



Determine the coordinates of P .

(9)
(Total 12 marks)

Q8.

The coordinates of the points A and B are $(3, -2, 4)$ and $(6, 0, 3)$ respectively.

$$\mathbf{r} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

The line l_1 has equation

- (a) (i) Find the vector $\overrightarrow{AB}$.

(2)

- (ii) Calculate the acute angle between $\overrightarrow{AB}$ and the line l_1 , giving your answer to the nearest 0.1° .

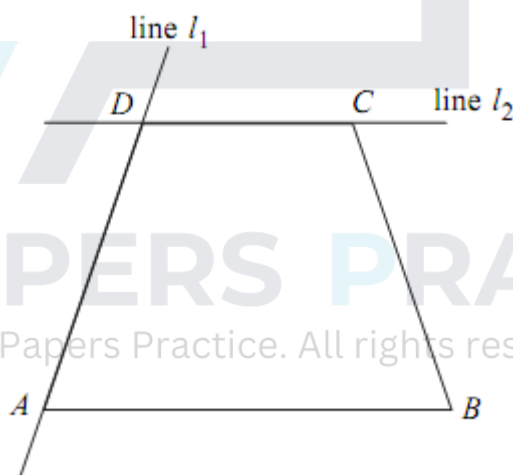
(4)

- (b) The point D lies on l_1 where $\lambda = 2$. The line l_2 passes through D and is parallel to AB .

- (i) Find a vector equation of line l_2 with parameter μ .

(2)

- (ii) The diagram shows a symmetrical trapezium $ABCD$, with angle DAB equal to angle ABC .



The point C lies on line l_2 . The length of AD is equal to the length of BC .

Find the coordinates of C .

(6)

(Total 14 marks)

Q9.

The points A and B have coordinates $(5, 1, -2)$ and $(4, -1, 3)$ respectively.

$$\mathbf{r} = \begin{bmatrix} -8 \\ 5 \\ -6 \end{bmatrix} + \mu \begin{bmatrix} 5 \\ 0 \\ -2 \end{bmatrix}$$

The line l has equation

- (a) Find a vector equation of the line that passes through A and B . (3)
- (b) (i) Show that the line that passes through A and B intersects the line l , and find the coordinates of the point of intersection, P . (4)
- (ii) The point C lies on l such that triangle PBC has a right angle at B . Find the coordinates of C . (5)
- (Total 12 marks)

Q10.

The points A and B have coordinates $(2, 1, -1)$ and $(3, 1, -2)$ respectively. The angle OBA is θ , where O is the origin.

- (a) (i) Find the vector $\overrightarrow{AB}$. (2)
- (ii) Show that $\cos \theta = \frac{5}{2\sqrt{7}}$. (4)
- (b) The point C is such that $\overrightarrow{OC} = 2\overrightarrow{OB}$. The line l is parallel to $\overrightarrow{AB}$ and passes through the point C . Find a vector equation of l . (2)
- (c) The point D lies on l such that angle $ODC = 90^\circ$. Find the coordinates of D . (4)
- (Total 12 marks)

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Q11.

The points A and B lie on the line l_1 and have coordinates $(2, 5, 1)$ and $(4, 1, -2)$ respectively.

- (a) (i) Find the vector $\overrightarrow{AB}$. (2)
- (ii) Find a vector equation of the line l_1 , with parameter λ . (1)
- (b) The line l_2 has equation $\mathbf{r} = \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$.
- (i) Show that the point $P(-2, -3, 5)$ lies on l_2 . (2)

- (ii) The point Q lies on l_1 and is such that PQ is perpendicular to l_2 . Find the coordinates of Q .

(6)

(Total 11 marks)

Q12.

The coordinates of the points A and B are $(3, -2, 1)$ and $(5, 3, 0)$ respectively.

The line l has equation $\mathbf{r} = \begin{bmatrix} 5 \\ 3 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$.

- (a) Find the distance between A and B . (2)
- (b) Find the acute angle between the lines AB and l . Give your answer to the nearest degree. (5)
- (c) The points B and C lie on l such that the distance AC is equal to the distance AB . Find the coordinates of C . (5)

(Total 12 marks)

Q13.

The points A , B and C have coordinates $(3, -2, 4)$, $(5, 4, 0)$ and $(11, 6, -4)$ respectively.

- (a) (i) Find the vector $\overrightarrow{BA}$. (2)
- (ii) Show that the size of angle ABC is $\cos^{-1}\left(-\frac{5}{7}\right)$. (5)

(b) The line l has equation $\mathbf{r} = \begin{bmatrix} 8 \\ -3 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$

- (i) Verify that C lies on l . (2)
- (ii) Show that AB is parallel to l . (1)
- (c) The quadrilateral $ABCD$ is a parallelogram. Find the coordinates of D . (3)

(Total 13 marks)

Q14.

The lines l_1 and l_2 have equations $\mathbf{r} = \begin{bmatrix} 8 \\ 6 \\ -9 \end{bmatrix} + \lambda \begin{bmatrix} 3 \\ -3 \\ -1 \end{bmatrix}$ and $\mathbf{r} = \begin{bmatrix} -4 \\ 0 \\ 11 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ respectively.

- (a) Show that l_1 and l_2 are perpendicular. (2)
- (b) Show that l_1 and l_2 intersect and find the coordinates of the point of intersection, P . (5)
- (c) The point $A (-4, 0, 11)$ lies on l_2 . The point B on l_1 is such that $AP = BP$. Find the length of AB . (4)

(Total 11 marks)

Q15.

The line l has equation $\mathbf{r} = \begin{bmatrix} 3 \\ 26 \\ -15 \end{bmatrix} + \lambda \begin{bmatrix} 8 \\ -4 \\ 1 \end{bmatrix}$.

- (a) Show that the point $P (-29, 42, -19)$ lies on l . (1)
- (b) Find the acute angle between l and the z -axis. (1)
- (c) The plane Π has cartesian equation $3x - 4y + 5z = 100$.
 - (i) Write down a normal vector to Π . (1)
 - (ii) Find the acute angle between l and this normal vector. (4)
- (d) Find the position vector of the point Q where l meets Π . (4)
- (e) Determine the shortest distance from P to Π . (3)

(Total 14 marks)

Q16.

The quadrilateral $ABCD$ has vertices $A (2, 1, 3)$, $B (6, 5, 3)$, $C (6, 1, -1)$ and $D (2, -3, -1)$.

The line l_1 has vector equation $\mathbf{r} = \begin{bmatrix} 6 \\ 1 \\ -1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$.

- (a) (i) Find the vector $\vec{AB}$. (2)
- (ii) Show that the line AB is parallel to l_1 . (1)
- (iii) Verify that D lies on l_1 . (2)
- (b) The line l_2 passes through $D(2, -3, -1)$ and $M(4, 1, 1)$.
- (i) Find the vector equation of l_2 . (2)
- (ii) Find the angle between l_2 and AC . (3)
- (Total 10 marks)**

Q17.

The points A and B have coordinates $(2, 4, 1)$ and $(3, 2, -1)$ respectively. The point C is such that $\vec{OC} = 2\vec{OB}$, where O is the origin.

- (a) Find the vectors:
- (i) $\vec{OC}$; (1)
- (ii) $\vec{AB}$. (2)
- (b) (i) Show that the distance between the points A and C is 5. (2)
- (ii) Find the size of angle BAC , giving your answer to the nearest degree. (4)
- (c) The point $P(\alpha, \beta, \gamma)$ is such that BP is perpendicular to AC .
- Show that $4\alpha - 3\gamma = 15$. (3)
- (Total 12 marks)**

Q18.

The points A and B have coordinates $(1, 4, 2)$ and $(2, -1, 3)$ respectively.



The line l has equation $\mathbf{r} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

(a) Show that the distance between the points A and B is $3\sqrt{3}$.

(2)

(b) The line AB makes an acute angle θ with l . Show that $\cos \theta = \frac{7}{9}$.

(3)

(c) The point P on the line l is where $\lambda = p$.

(i) Show that

$$\vec{AP} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 7 + 3p$$

(4)

(ii) Hence find the coordinates of the foot of the perpendicular from the point A to the line l .

(3)

(Total 12 marks)